



Performance at a Glance

62 / 65

TOTAL SCORE

95.4 %

PERCENTAGE

A+

GRADE

3

MARKS LOST

Section Breakdown

Section	Description	Score	%	Marks Lost
A	MCQs (12 × 1) — Dynamics, equilibrium, Nature of Science	12 / 12	100 %	0
B	Short Answers (11 × 3) — Force, Newton's laws, momentum, equilibrium, friction, terminal velocity	30.5 / 33	92.4 %	2.5
C	Long Answers (4 × 5) — Newton's laws + limitations, impulse-momentum derivation, equilibrium & CG, centripetal force / orbital speed	19.5 / 20	97.5 %	0.5
Total	Physics — exam-02-ch3-4-9 (Version 2)	62 / 65	95.4 %	3

Detailed View

Physics — exam-02-ch3-4-9

62 / 65 (95.4 %) Grade A+

Unit 3 (Dynamics-I) · Unit 4 (Dynamics-II) · Unit 9 (Nature of Science) · 10 April 2026

12 / 12

Section A · 100 %

30.5 / 33

Section B · 92.4 %

19.5 / 20

Section C · 97.5 %

62 / 65

Total · 95.4 %

Strengths

- Perfect MCQs (12 / 12) — strong dynamics and equilibrium concepts
- Flawless numerical work: momentum, hammer throw, Earth's orbital speed
- Full first-principles derivation of impulse-momentum and $v^2 = GM/r$
- Newton's three laws stated with math forms and clear examples
- Principle of moments verified correctly ($0.8 \text{ N}\cdot\text{m} = 0.8 \text{ N}\cdot\text{m}$)
- Section C average 97.5 % — strong long-answer discipline

Areas to Improve

- Arithmetic check (Q.4 b·i):** $9600 / 5 = 1920 \text{ N}$, wrote 1020 — **−0.5**
- Terminal velocity (Q.2 ix):** missed the drag-equals-weight balance — **−1.5**
- Equilibrium definition (Q.2 vii):** didn't define equilibrium itself — **−0.5**
- Conservation of momentum form (Q.2 v):** use $m_1v_{1i} + m_2v_{2i} = m_1v_{1f} + m_2v_{2f}$ — **−0.5**
- Unit slips — "0.25 m/s" vs m/s^2 ; "0.75 $\text{N}\cdot\text{m}$ " vs m (no deduction this time)

Comparison vs Previous Physics Exam

Exam	Date	Section A	Section B	Section C	Total	%
exam-01-ch1-2 (Measurement, Kinematics)	7 Apr 2026	12 / 12	32.5 / 33	19 / 20	63.5 / 65	97.7 %

exam-02-ch3-4-9 (Dynamics, Nature of Science)	10 Apr 2026	12 / 12	30.5 / 33	19.5 / 20	62 / 65	95.4 %
Δ vs previous		— same	▼ -2.0	▲ +0.5	▼ -1.5	▼ -2.3 pts

Key Takeaways

- **MCQ mastery is consistent** — 12 / 12 across both Physics exams (24 / 24).
- **Section C improved** — 95 % → 97.5 %, showing better long-answer discipline.
- **Section B dipped** — the drop is concentrated in two specific gaps (terminal velocity concept, conservation of momentum form) rather than a broad weakness.
- **Conceptual recall** for Newton's laws, impulse-momentum derivation, and centripetal / orbital motion is excellent.

Things to Fix Before the Next Exam

- **Memorise the canonical sentence** for terminal velocity: *"Drag force grows with speed until it equals the weight; then $F_{net} = 0$ and $a = 0$, so velocity stays constant."*
- **Always define** the umbrella term before listing conditions (e.g. "Equilibrium: net force AND net torque are zero" before stating the two conditions).
- **Two-pass arithmetic check** on every numerical — recompute the final division step before writing the answer.
- **Conservation of momentum two-body form:** $m_1v_{1i} + m_2v_{2i} = m_1v_{1f} + m_2v_{2f}$.
- **Unit hygiene:** always write m/s^2 (not m/s) for acceleration, and m (not $N\cdot m$) for distance.